

# Amanda Stoffers

EXTRAGALACTIC ASTROPHYSICS & BAYESIAN INFERENCE METHODS

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## Research Profile

I work at the interface of galaxy evolution and statistical methodology. My scientific focus is the evolution of galaxies in the first billion years: what regulated their ionising output, and how did it evolve with star formation and feedback? I study the coupling between star formation, ionising photon production and escape that sets the galaxy contribution to Reionisation. Methodologically, I develop fast and flexible inference tools designed to scale to entire surveys with a focus on high redshift galaxies, including the differentiable, GPU-native spectral energy distribution (SED) fitting software CERIDWEN described below.

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## Education

- 2023 – Present **PhD in Physics**,  
*University of Cambridge*, UK  
Kavli Institute for Cosmology and Cavendish Laboratory, Galaxy Formation and Evolution Group. Supervisor: Dr Sandro Tacchella.
- 2023 – 2024 **Graduate Training Programme, Data Intensive Science**,  
*University of Cambridge*, UK  
Advanced coursework in statistics, machine learning and scientific computing.
- 2022 – 2023 **MASt in Physics**,  
*University of Cambridge*, Cambridge, UK  
Research project on the escape fraction of ionising photons, awarded First Class. Overall result: Upper Second.
- 2018 – 2022 **BSc in Physics**, *Technische Universität Berlin*, Berlin, Germany • 1.3  
Thesis: *Quantum theory of electron-phonon interaction in gold nanoparticles* (1.0).  
German scale, where 1.0 is the highest attainable grade.

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## Research Software

- 2024 – Present **Ceridwen**, *Author/Lead developer*, [github.com/Espe13/ceridwen](https://github.com/Espe13/ceridwen)  
A GPU-native, end-to-end differentiable SED-fitting framework in JAX, covering the full forward model from simple stellar populations through nebular emission and dust to projections onto the observed frame. Production inference uses a GPU nested slice sampler, with Hamiltonian Monte Carlo and variational inference available for the same model; supports non-parametric star formation histories of order one hundred age bins with jointly sampled chemical enrichment. A median of 134 times faster than CPU PROSPECTOR on high redshift galaxies. MIT licensed.
- 2025 – Present **sedpy\_jax**, *Author*, [github.com/Espe13/sedpy\\_jax](https://github.com/Espe13/sedpy_jax)  
A JAX reimplementation of **sedpy** (B. D. Johnson): GPU-ready, differentiable filter handling and broadband photometric projection: transmission curves, AB and Vega magnitudes, flux-conserving rebinning, air–vacuum conversion, and effective, pivot and zero-point wavelengths across many filters at once. MIT licensed.

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## Major Collaborations

- 2023 – Present **JADES**, *JWST Advanced Deep Extragalactic Survey*  
Leading author of two collaboration papers in preparation. Leading the construction of a value-added catalogue of inferred physical properties for the JADES galaxies with both NIRCам photometry and NIRSpec spectroscopy, fitted jointly with CERIDWEN for use across the collaboration.
- 2026 – Present **LAPIS**, *Large Area Parallel Imaging Survey*  
Co-Investigator on a JWST pure-parallel survey covering  $> 1000 \text{ arcmin}^2$  across  $> 100$  independent sightlines to constrain the abundance and properties of galaxies at  $z > 16$ .

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## Publications

### First author

- 2026 **The Challenge in Illuminating the Invisible: Constraining LyC Escape with Bayesian Modelling and Symbolic Regression**  
Stoffers, A., Tacchella, S., Simmonds, C., Johnson, B. D., & Maiolino, R.  
*Monthly Notices of the Royal Astronomical Society*, 549, *stag984* (*arXiv:2511.02908*)  
[\[ADS\]](#)

- 2026 **CERIDWEN: Fast and Flexible GPU-Accelerated Stellar Population Inference**  
Stoffers, A., Tacchella, S., & Johnson, B. D.  
*Submitted to Monthly Notices of the Royal Astronomical Society*  
(*arXiv:2609.30145*, *doi:10.48550/arXiv.2609.30145*) [\[ADS\]](#)

### First author in prep

- expected October 2026 **JADES: Inferred Ionizing Photon Escape Fractions and the Ionizing Budget of Galaxies**  
Stoffers, A., Tacchella, S., Johnson, B. D., Bunker, A., Robertson, B., Simmonds, C., Whitler, S., Witstok, J.

- expected October 2026 **JADES: the Role of Star-formation Variability in Shaping the UV Luminosity Function at  $z \sim 3 - 9$**   
Stoffers, A., Tacchella, S., Johnson, B. D., Duan, Q., Robertson, B.

### Co-authored

- 2026 **From THESAN-ZOOM to JWST: Predicting ionizing photon escape and the rise of UV-bright reionization sources**  
Summerfield, Z., McClymont, W., Tacchella, S., Smith, A., Kannan, R., et al., incl. Stoffers, A. (17 authors; 15th)  
*Submitted to Monthly Notices of the Royal Astronomical Society* (*arXiv:2606.14671*) [\[ADS\]](#)  
Contribution: discussion of the comparison between simulated and observed escape fractions; advice on the manuscript.

- 2026 **An (in)complete NIRSpect census of Balmer absorption in Type 1 AGN: radiation-driven outflows in little red dots, quasars and variable stars**  
Juodžbalis, I., Ji, X., D'Eugenio, F., Scholtz, J., Maiolino, R., Stoffers, A., et al. (20 authors; 6th)  
*Submitted* (*arXiv:2607.19057*) [\[ADS\]](#)  
Contribution: statistical model for the parameter-recovery test; completeness correction for the incidence rate of Balmer absorption.

- 2025 **Bursting at the seams: the star-forming main sequence and its scatter at  $z = 3-9$  using NIRCам photometry from JADES**  
Simmonds, C., Tacchella, S., McClymont, W., Curtis-Lake, E., D'Eugenio, F., et al., incl. Stoffers, A. (27 authors; 11th)  
*Submitted* (*arXiv:2508.04410*) [\[ADS\]](#)  
Contribution: parameter-recovery tests for the inferred star-formation histories; injection tests for the recovery of the 10- and 100-Myr SFR indicators.

- 2024 **Ionizing properties of galaxies in JADES for a stellar mass complete sample: resolving the cosmic ionizing photon budget crisis at the Epoch of Reionization**  
Simmonds, C., Tacchella, S., Hainline, K., Johnson, B. D., Puskás, D., et al., incl. Stoffers, A. (29 authors; 23rd)  
*Monthly Notices of the Royal Astronomical Society*, 535(4), 2998–3019 [\[ADS\]](#)  
Contribution: parameter-recovery and injection tests for the inferred star-formation histories.

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## Invited Talks

- 2026 **Invited seminar**, Friday Late Afternoon Talks, UCL, London, UK  
*Differentiable Galaxies: What Gradient-Based Inference Changes About SED Fitting*
- 2025 **Invited talk**, JADES Science Focus Meeting, Santa Cruz, CA, USA  
*CERIDWEN: Fitting JADES Star Formation Histories at Survey Scale on GPUs*

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## Contributed Talks and Presentations

- 2026 **Contributed talk**, Kavli Symposium 2026, Kavli Institute for Cosmology, Cambridge, UK  
*The Rhythm of Star Formation: SF Variability from  $z = 3-9$  in JADES*
- 2026 **Contributed talk**, Simulation Based Inference for Galaxy Evolution, Cambridge, UK  
*Building Better SBI from Better Simulators: Differentiable SED Modelling with CERIDWEN*
- 2025 **Contributed talk**, LyC Escape from Galactic Labyrinths, Crete, Greece  
*Inferring Lyman Continuum Escape Fractions via SED Modelling*
- 2025 **Contributed talk**, Kaba Kada: Exploring the First Billion Years of the Universe, Port Douglas, Australia  
*The Challenge in Illuminating the Invisible: Constraining LyC Escape with Bayesian Modelling and Symbolic Regression*
- 2025 **Workshop presentation**, IAIFI Summer School and Workshop, Harvard University, Boston, MA, USA  
*Contextual Bandits for Stroke Treatment Analysis*
- 2024 **Workshop presentation**, IAIFI Summer School and Workshop, MIT, Boston, MA, USA  
*Anomaly Detection via Latent Space Evaluations*
- 2024 **Contributed talk**, National Astronomy Meeting (NAM), Hull, UK  
*Lyman Continuum Escape Fractions of High-Redshift Galaxies*

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## Grants, Awards and Scholarships

- 2026 **DiRAC Seedcorn Award**, STFC DiRAC High Performance Computing Facility, Tursa GPU cluster, Edinburgh  
Principal Investigator. 1000 hours GPU time for CERIDWEN fits of the JADES sample.
- 2023 – 2027 **STFC CDT in Data Intensive Science studentship**, grant ST/W006812/1, Cambridge, UK  
Full fees and stipend; partly co-funded by the UKRI Frontier Research grant RISEandFALL.
- 2025 **ESA-funded research visit**, JADES Science Focus Meeting, Santa Cruz, CA, USA
- 2025 **Postgraduate Research Fund**, Jesus College, Cambridge  
Travel award to present at *Kaba Kada: Exploring the First Billion Years of the Universe*, Port Douglas, Australia.
- 2025 **Research Culture Grant**, Kavli Institute for Cosmology, Cambridge, UK  
Awarded to develop and run the Astro Career Talk Series.
- 2025 **Best Talk and Presentation**, Graduate Research Conference, Jesus College, Cambridge, UK
- 2021 – 2023 **Studienstiftung des deutschen Volkes**, German Academic Scholarship Foundation  
Germany's national academic scholarship, awarded to roughly the top 0.5% of students.
- 2021 **Enable AI Scholarship**, Data science and machine learning training
- Earlier KMK Scholarship, exchange programme, Trentino, Italy (2017); Mathematics Abitur Award, Deutsche Mathematiker-Vereinigung (2018); 1st place, Berlin regional Mathematics Olympiad (2014).

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## Teaching and Supervision

- 2023 – 2025 **Supervisor**, *Relativistic Astrophysics and Cosmology*, Cavendish Laboratory, University of Cambridge  
Small-group undergraduate supervisions, run in each of the 2023, 2024 and 2025 academic years.
- 2023 **Supervisor**, *Mathematics for Natural Sciences, Part IA*, Trinity Hall, University of Cambridge  
First-year small-group supervisions.

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## Professional Service and Leadership

- 2026 **Local Organising Committee**, *Hubble and Beyond: Morphology and Kinematics of Early Galaxies*, Kavli Institute for Cosmology, Cambridge
- 2024 – 2025 **Committee Member**, *Astro Career Talk Series*, Kavli Institute for Cosmology, Cambridge  
Helped develop a departmental series of interviews with academic and industry researchers into its current form; secured a Research Culture Grant to fund it.
- 2024 – 2025 **Committee Member**, *Cambridge University Women in Physics Society*
- 2024 **Local Organising Committee**, *Star Formation 2024*, Kavli Institute for Cosmology, Cambridge
- 2022 – 2023 **Student Representative**, *MASt Physics Programme*, University of Cambridge
- 2021 – 2025 **Board Member**, *In.Ph.A.S.E. Student Association*  
Organised the student colloquium programme.
- 2024 – Present **Jesus College Boat Club**, University of Cambridge  
W1 Captain (2024–2025), leading and coaching the college's senior women's crew; Committee Member (2026–present).

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## Outreach and Public Engagement

- 2018 – 2023 **Presenter and Event Host**, *Planetarium Berlin*, Berlin, Germany  
Public astronomy talks and show presentation for general audiences.
- 2019 – 2022 **Volunteer Instructor**, *Observatories e.V.*, Berlin, Germany  
Ran observing sessions and hands-on experiments for public observatory programmes.
- 2021 – 2022 **Student Laboratory Tutor**, *DESY Zeuthen*, Zeuthen, Germany  
Supervised school students running particle-physics experiments.

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## Other Professional Experience

- 2023 **Quantum Software Engineer**, *Diehl Defence*, Überlingen, Germany  
Implemented quantum algorithms and developed the associated mathematical models.
- 2021 – 2022 **Educational Assistant**, *Grunewald Primary School*, Berlin, Germany

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## Technical Skills

- Inference Nested sampling, Hamiltonian Monte Carlo, variational inference, hierarchical Bayesian modelling, symbolic regression, automatic differentiation
- Computing Python (JAX, NumPyro, PROSPECTOR, Qiskit, QuTiP), C/C++, GPU programming, Git, Linux, high-performance computing, L<sup>A</sup>T<sub>E</sub>X
- Languages German (native), English (near-native), French (B2), Italian (A2), Spanish (A1)